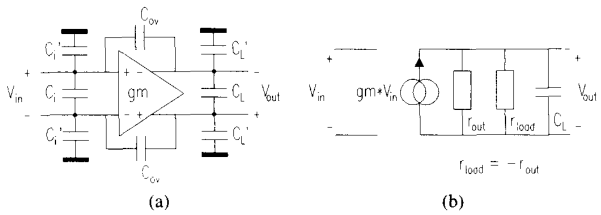

Fig. 1. (a) A transconductor without internal nodes will have no parasitic poles or zeros and will be therefore of infinite bandwidth. (b) Loading the output of an integrator with a negative load resistance makes a infinite dc gain possible, without requiring internal nodes.

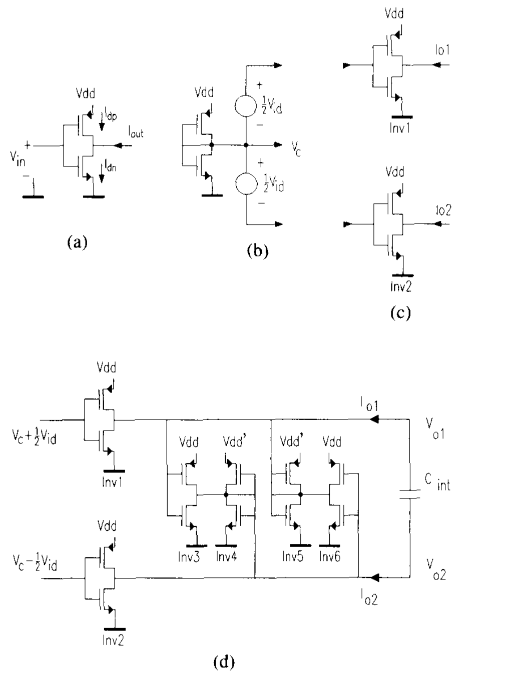

Fig. 2. (a) Single inverter. (b) Generation of the common-mode voltage level V_c. (c) Two balanced inverters performing linear V-to-I conversion if driven by the circuit of Fig. 2(b). (d) The complete transconductance element.

theoretically infinite dc gain and infinite bandwidth. As a result, the integrator quality factor will also be infinite.

In this paper, first a tranconductor circuit with an excellent high-frequency behavior is described (Section II). Then, for demonstration, a third-order elliptic filter with a cutoff frequency tunable up to 98 MHz is described (Section III). For this transconductor a Q-tuning circuit with high-speed potential (Section IV) and a supply voltage buffer (Section V) are also presented. Finally, the experimental results of the circuits are discussed (Section VI).

II. Transconductor

In this section, first the linear V-to-I conversion of the transconductor is described and then the common-mode control, dc-gain enhancement, bandwidth, distortion, and noise are discussed.

A. V–I Conversion

The transconductor [9] is based upon the well-known CMOS inverter. This CMOS inverter has no internal nodes and has a good linearity in V–I conversion if the β factors of the n-channel and p-channel transistors are perfectly matched. Consider first the inverter of Fig. 2(a). If the drain currents of an n- and a p-channel MOS transistor in saturation are written as

$$I_{dn} = \frac{\beta_n}{2}(V_{gsn} - V_{tn})^2, \quad \text{with } \beta_n = \frac{\mu_n C_{ox} W_n}{L_n} \tag{1a}$$

$$I_{dp} = \frac{\beta_p}{2}(V_{gsp} - V_{tp})^2, \quad \text{with } \beta_p = \frac{\mu_p C_{ox} W_p}{L_p} \tag{1b}$$

then the output current of the single inverter can be written as

$$I_{\text{out}} = I_{dn} - I_{dp} = a(V_{\text{in}} - V_{tn})^2 + b \cdot V_{\text{in}} + c \tag{2}$$

with

$$a = \tfrac{1}{2}(\beta_n - \beta_p) \tag{2a}$$

$$b = \beta_p(V_{dd} - V_{tn} + V_{tp}) \tag{2b}$$

$$c = \tfrac{1}{2}\beta_p(V_{tn}^2 - (V_{dd} + V_{tp})^2). \tag{2c}$$

All devices are assumed to operate in strong inversion and in saturation. If $\beta_n \neq \beta_p$, i.e., $a \neq 0$, the V-to-I conversion will not be linear. The error is in fact a square-law term, that can be canceled if a balanced structure is used.

The output current is zero when $V_{\text{in}} = V_c$ (see Fig. 2(b)), with

$$V_c = \frac{V_{dd} - V_{tn} + V_{tp}}{1 + \sqrt{\dfrac{\beta_n}{\beta_p}}} + V_{tn}. \tag{3}$$

Note that for $\beta_p = \beta_n$ and $V_{tn} = -V_{tp}$, then $V_c = 1/2\, V_{dd}$ as can be easily verified.

Fig. 2(c) shows the balanced version of the circuit of Fig. 2(a). The two matched inverters Inv1 and Inv2 are driven by a differential input voltage V_{id}, balanced around the common-mode voltage level V_c (see (3)). The output currents I_{o1} and I_{o2} can be calculated, and subtraction results in the differential output current I_{od}:

$$I_{o1} = a(V_c - V_{tn} + \tfrac{1}{2}V_{id})^2 + b(V_c + \tfrac{1}{2}V_{id}) + c$$

$$I_{o2} = a(V_c - V_{tn} - \tfrac{1}{2}V_{id})^2 + b(V_c - \tfrac{1}{2}V_{id}) + c$$

$$I_{o1} - I_{o2} = a((V_c - V_{tn} + \tfrac{1}{2}V_{id})^2 - (V_c - V_{tn} - \tfrac{1}{2}V_{id})^2) + bV_{id}$$

or

$$I_{o1} - I_{o2} = V_{id}(b + 2a(V_c - V_{tn})) = V_{id}(\beta_p(V_{dd} - V_c + V_{tp}) + \beta_n(V_c - V_{tn})). \tag{4}$$